



Blind Enhanced and Assistive Cap with Optical Navigation

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INTRODUCTION

- Assistive system designed for visually impaired users using AI and sensors
- Detects obstacles and provides real-time navigation support
- Enhances safety and independence through smart wearable technology



ABSTRACT

- User accesses system via mobile app
- App provides multiple safety alerts
- Backend + AI process data and generate alerts
- Camera enables real-time AI detection



OBJECTIVES

- Detect obstacles and classify objects in real time
- Provide navigation assistance through audio/haptic alerts
- Integrate fall detection and emergency alert system
- Ensure accuracy, portability, and ease of use



LITERATURE SURVEY



Papers	Authos(s)	Technical Findings	Gaps Identified
Safety Helmet Detection Based on Improved YOLOv8 (2024)	Xudong Song, Tiankai Zhang, Weiguo Yi	YOLOv8-based helmet detection	Single-task; no assistive features
Smart Helmet 5.0 for Industrial IoT Using AI (2020)	I. Campero-Jurado et al.	IoT sensors + CNN monitoring	No vision-based detection
Computer Vision Based Road Obstacle Detection (2025)	Hamza Assemlali et al.	Vision-based obstacle detection	Not wearable; no user alerts



Papers	Authos(s)	Technical Findings	Gaps Identified
Obstacle Detection for Visually Impaired Using IoT Sensors (2025)	Sunnia Ikram et al.	Ultrasonic/IR obstacle sensing	No object classification
Personal Voice Assistant Security and Privacy (2022)	Peng Cheng, Utz Roedig	Voice assistant security analysis	No navigation/awar eness
Smart Helmet Compliance Detection Using YOLO-M (2023)	Lili Wang et al.	YOLO-M helmet detection	Limited to single function
Visual Fall Detection for Assistive Living (2023)	Samyan Qayyum Wahla et al.	Video-based fall detection	No obstacle/navigati on support



Papers	Authos(s)	Technical Findings	Gaps Identified
Deep Learning Framework for Brazilian Coins (2023)	Debabrata Swain et al.	CNN-based classification of coin images	Static task only
Human-Centered AI in Smart Farming (2024)	Andreas Holzinger et al.	AI + IoT system for adaptive decision-making	Domain-specific
Omni-Directional Obstacle Detection (2020)	Xiangmo Zhao et al.	Depth camera-based 360° obstacle detection system	High cost; not wearable



EXISTING SYSTEM

- Existing assistive systems include smart glasses and sensor-based devices
- Provide features like obstacle detection, text recognition, and navigation
- Use camera-based AI or IoT sensors for assistance



EXISTING SYSTEM(Cont..)

- Most systems offer limited or single functionalities
- Features like detection, navigation, and alerts work independently
- High cost and complex design reduce accessibility
- Dependence on cloud processing causes latency
- Limited integration of AI vision and sensor data



PROBLEM STATEMENT

- Existing assistive devices are expensive and not easily accessible
- Limited integration of multiple functionalities in a single system
- Dependency on cloud-based AI leads to **delays in alert generation** during navigation.



PROBLEM STATEMENT(Cont..)

- Need for a system that supports continuous and multi-object detection
- visually impaired users require a system that can **adapt to diverse environments**, including indoor, outdoor, crowded, or low-light areas.
- Demand for a low-cost, portable, and user-friendly wearable solution
- Requirement useful AI models



PROPOSED SYSTEM

- Smart assistive cap integrating camera, sensors, and mobile application
- Uses AI models for multi-object detection (obstacles, text, currency)
- edge AI allows **on-device processing**, reducing latency and dependency on cloud or smartphone processing.



PROPOSED SYSTEM(Cont..)

- Combines multiple features: fall detection, alerts, and location tracking
- Uses smartphone + edge processing for faster response (reduced latency)
- All functionalities work together in a single system
- Low-cost, portable, and easy-to-use design



FEASIBILITY STUDY



TECHNICAL FEASIBILITY

- **hardware:** sensors (accelerometer), camera, microcontroller (esp32/arduino), edge ai modules are readily available.
- **software:** ai models (yolo, cnn) and mobile app platforms (react native) are mature and compatible with low-power devices.



OPERATIONAL FEASIBILITY

- **usability:** wearable cap design is portable and comfortable for visually impaired users.
- **real-time functionality:** AI ensures immediate obstacle detection and navigation alerts.
- **maintenance:** minimal; periodic calibration and software updates are sufficient.
- **adaptability:** can be used in indoor, outdoor, crowded, or low-light environments.



ECONOMICAL FEASIBILITY

- **hardware cost:** moderate; sensors, microcontrollers, and camera modules are affordable.
- **software cost:** minimal; open-source ai frameworks (tensorflow lite, yolo) reduce licensing expenses.
- **scalability:** potential for mass production; mobile app integration reduces additional costs.
- **long-term benefits:** improves safety and independence of visually impaired users, reducing reliance on expensive assistive technologies or caregivers.



REQUIREMENT SPECIFICATIONS



FUNCTIONAL REQUIREMENTS

- detect obstacles (stationary & moving)
- provide navigation guidance via audio/haptic alerts
- recognize obstacles and currency
- detect falls or collisions and trigger alerts
- process data on-device using edge ai
- communicate with mobile app via bluetooth/wifi

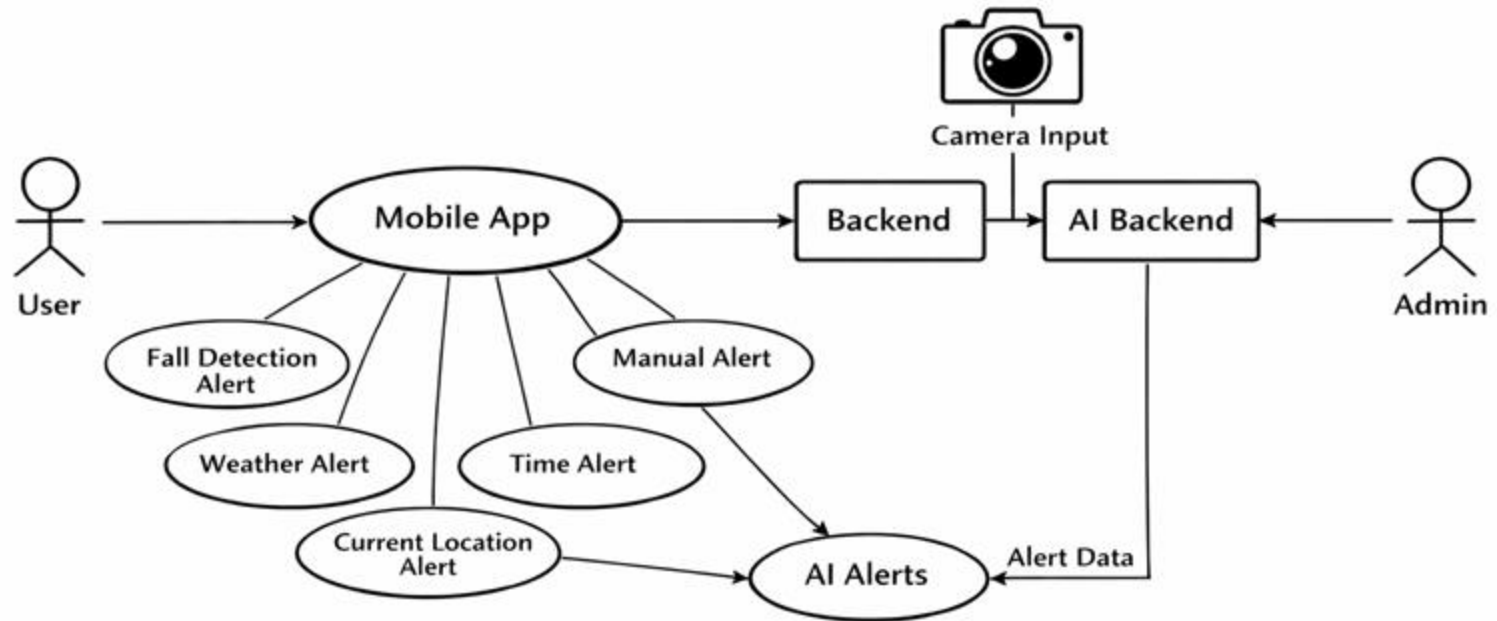


NON FUNCTIONAL REQUIREMENTS

- lightweight, comfortable, and easy to use
- reliable in various environments (indoor/outdoor, low-light)
- scalable for future enhancements
- portable and mostly independent of smartphone
- secure and privacy-protected data

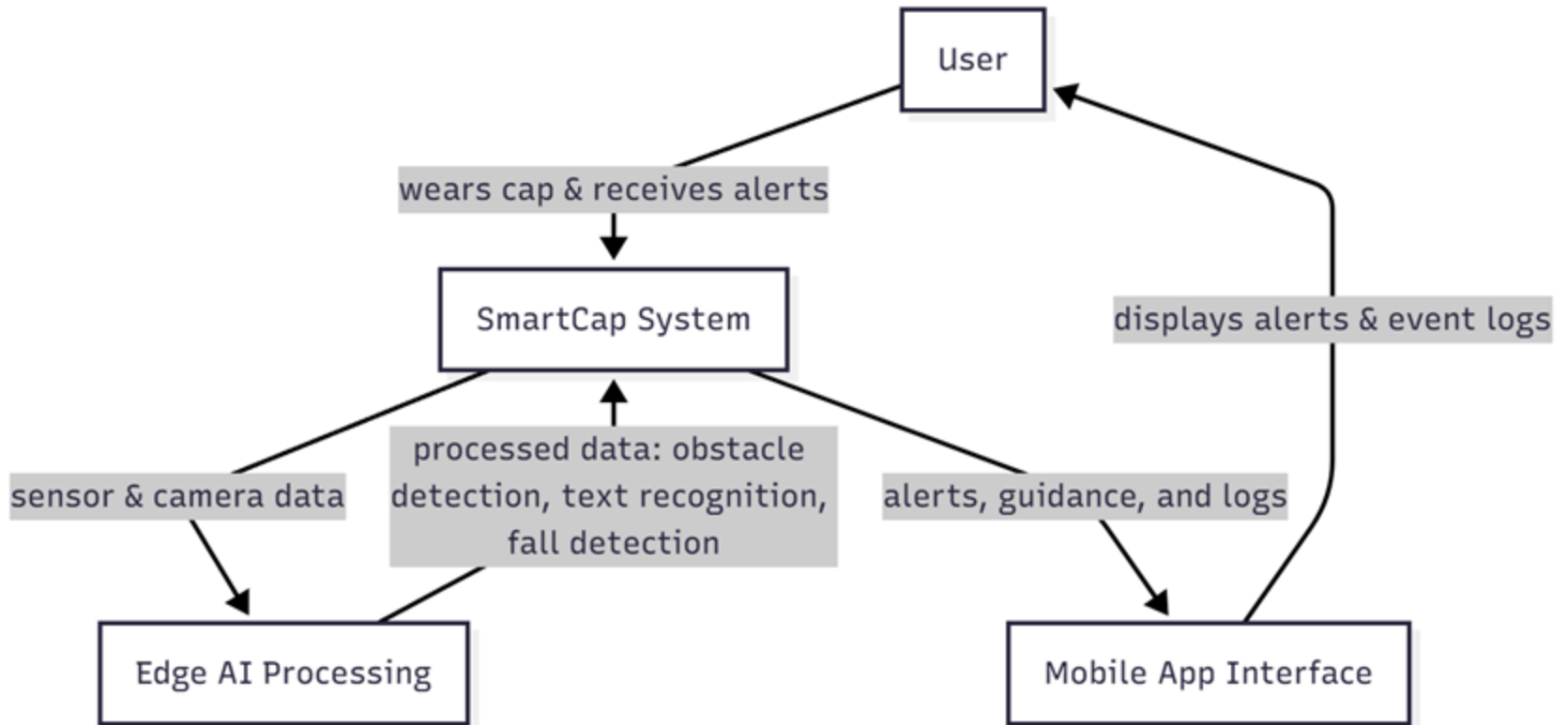


USE CASE DIAGRAM



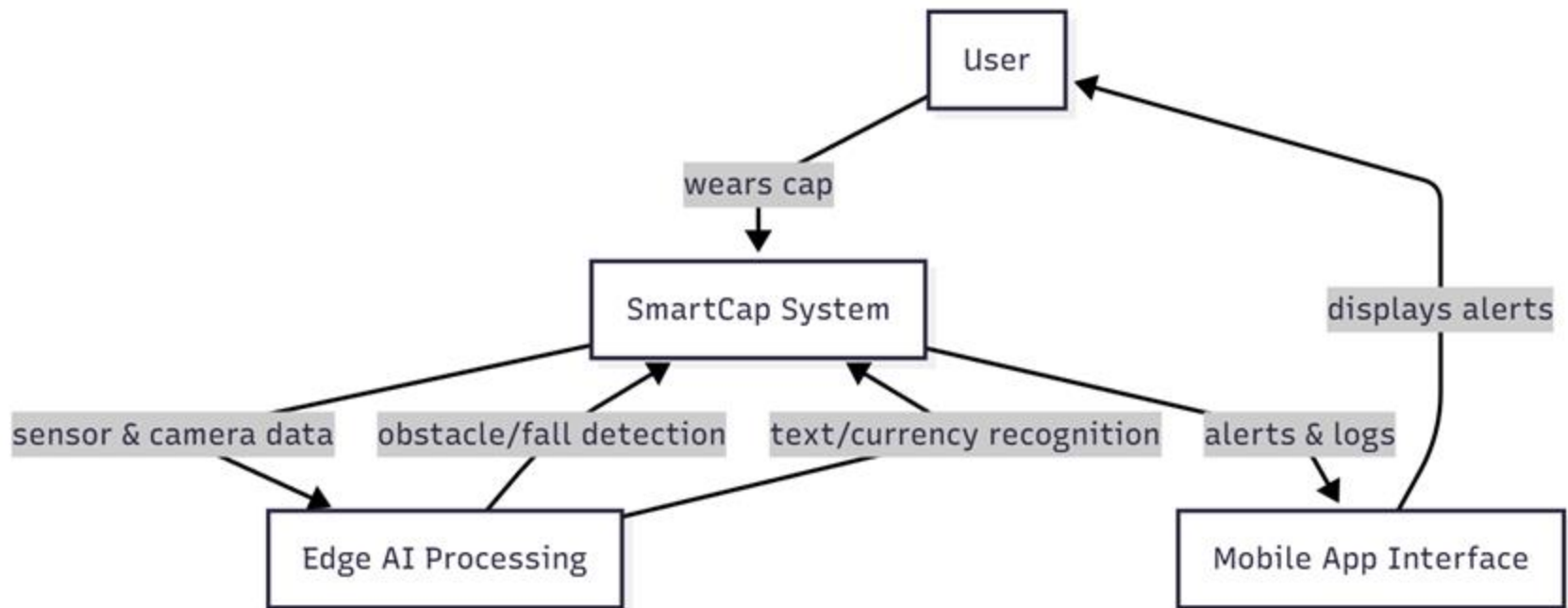
DATA FLOW DIAGRAM

LEVEL 0



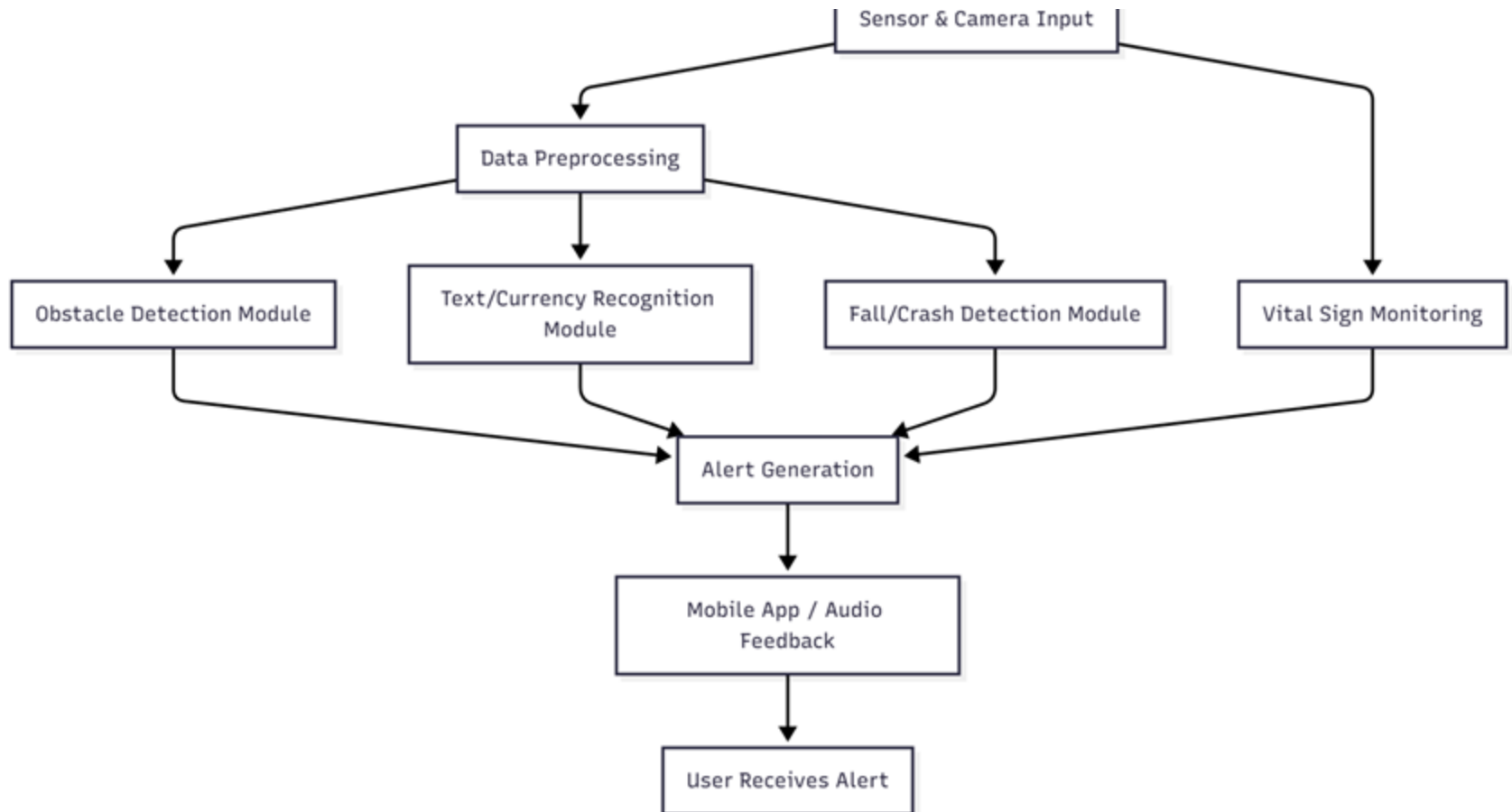
DATA FLOW DIAGRAM(Cont..)

LEVEL 1

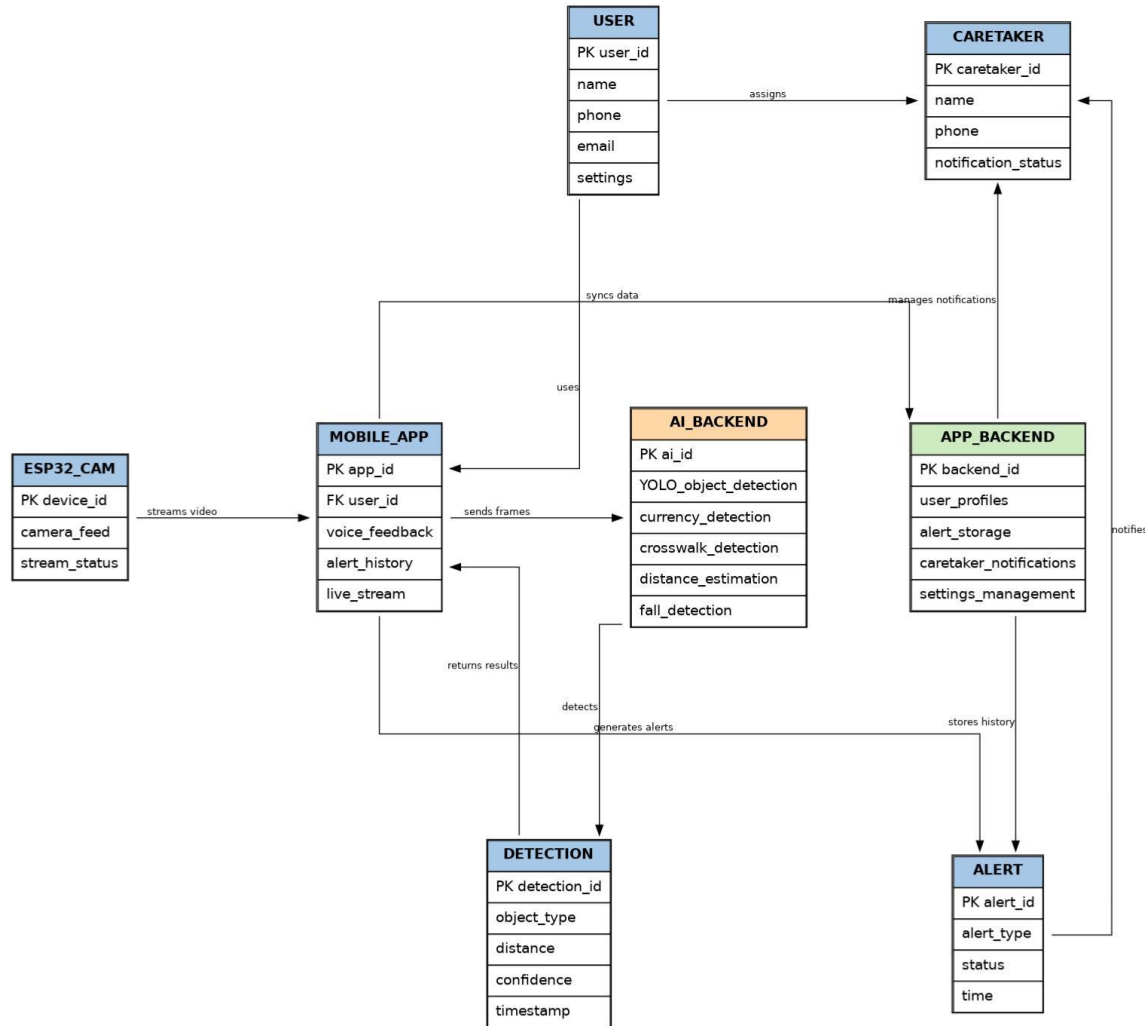


DATA FLOW DIAGRAM(Cont..)

LEVEL 2



ENTITY RELATIONSHIP DIAGRAM



RESULT AND DISCUSSION

- System successfully performs real-time obstacle detection and navigation assistance
- Accurate detection of multiple objects (obstacles, text, currency) using AI models
- Sensor + camera integration improves reliability of detection
- Fall detection and alert system works effectively with minimal delay
- Real-time alerts are generated through the mobile application

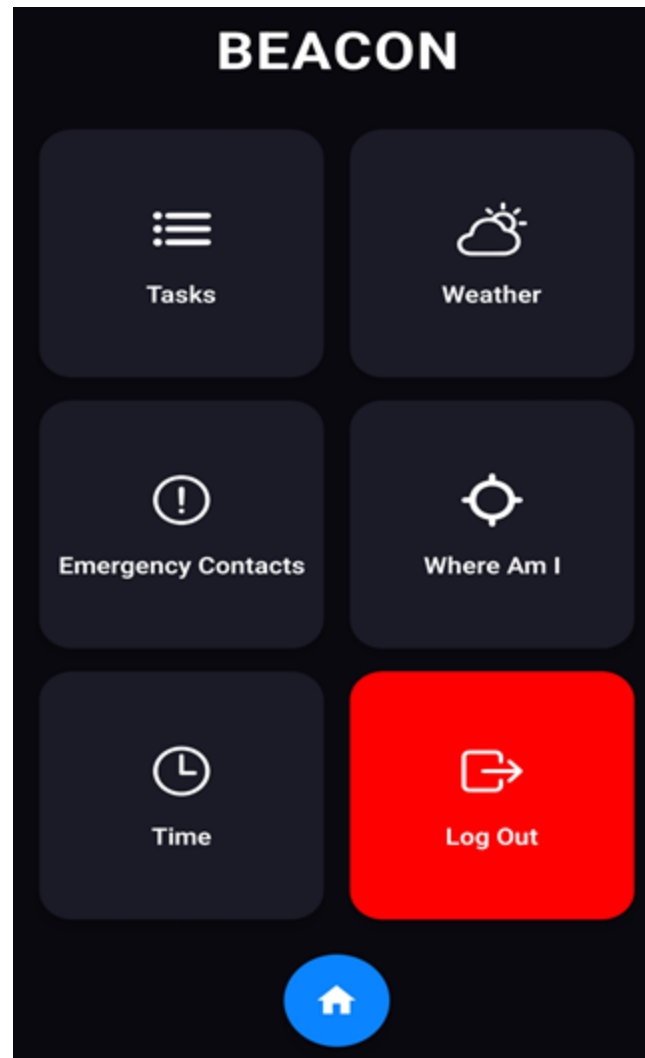


RESULT AND DISCUSSION

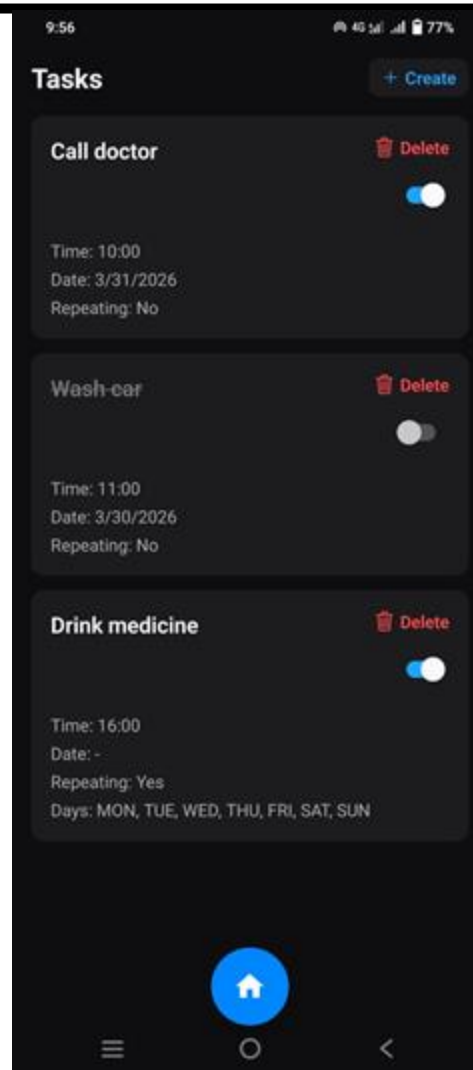
- Edge processing using smartphone ensures low latency and faster response
- System provides continuous monitoring and multi-functional assistance
- Portable and low-cost design makes it practical for real-world use
- Performance remains stable in different environments (indoor/outdoor)
- Overall system improves safety, independence, and user experience



RESULT AND DISCUSSION



RESULT AND DISCUSSION



RESULT AND DISCUSSION

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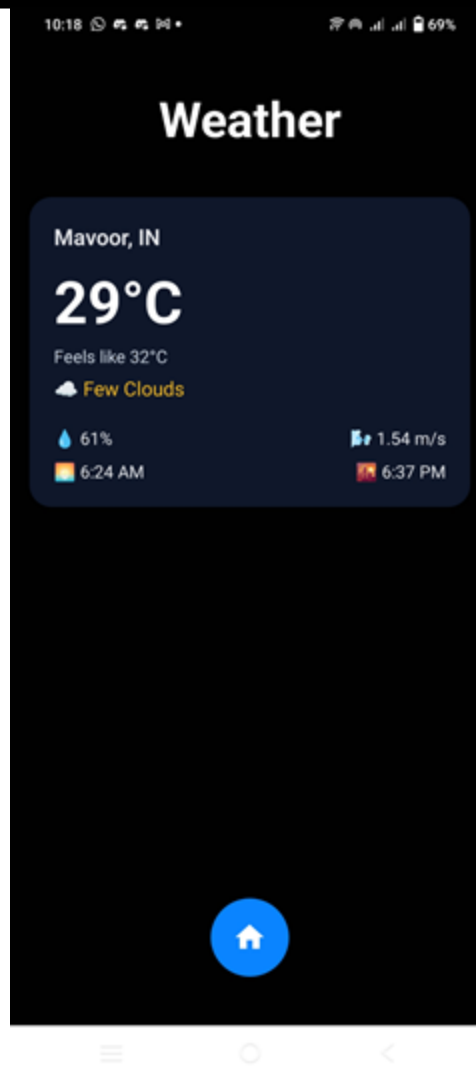
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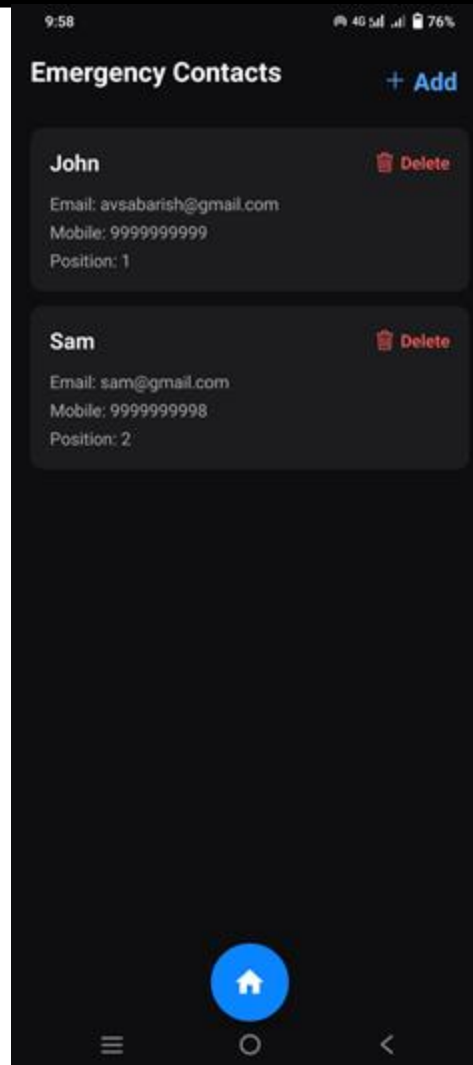
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RESULT AND DISCUSSION



RESULT AND DISCUSSION

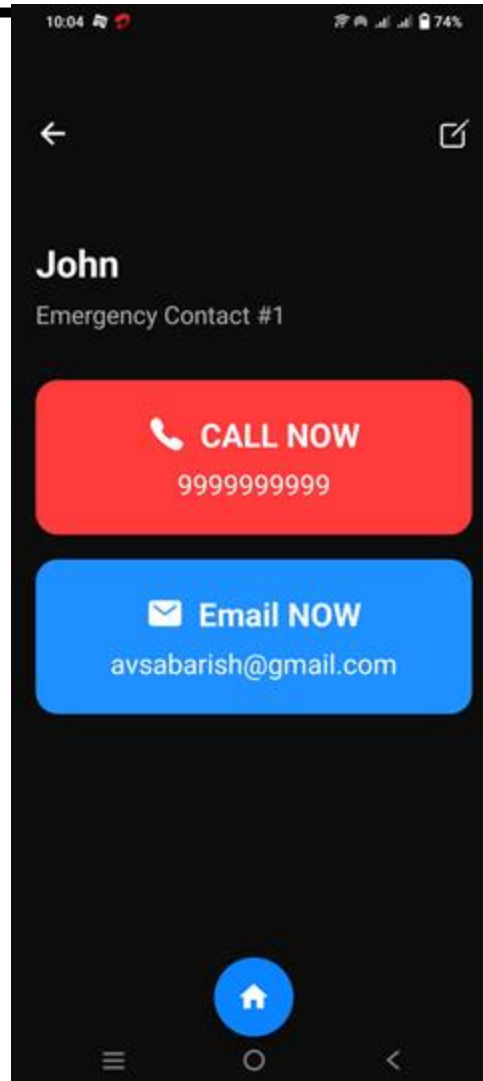


RESULT AND DISCUSSION

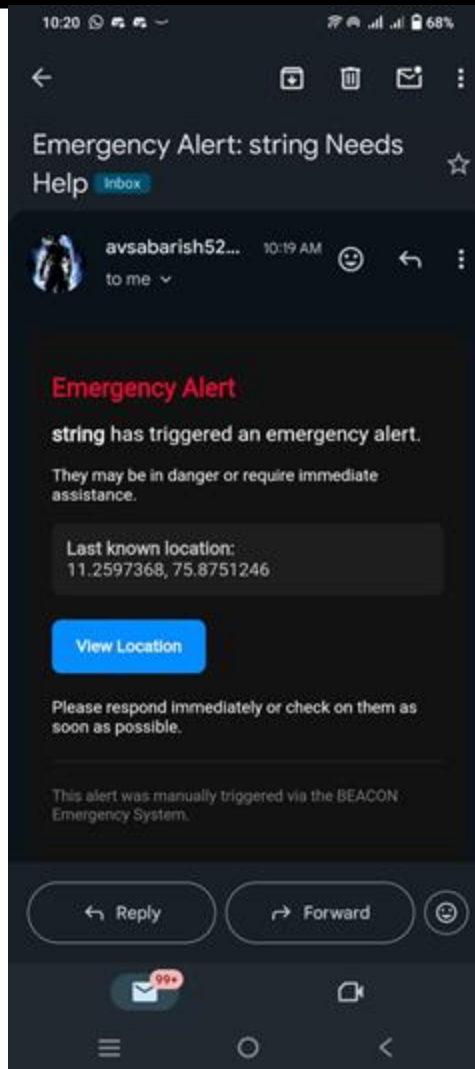
The screenshot shows a mobile application interface on a black background. At the top, the status bar displays the time 10:18, signal strength, and 69% battery. Below the status bar is a white back arrow icon. The title 'Add Emergency Contact' is centered in white. There are three input fields: 'Name', 'Email', and 'Mobile Number', each with a placeholder text of the same name. Below these fields is a blue 'Submit' button. At the bottom center is a blue circular button with a white house icon. The bottom of the screen shows the standard Android navigation bar with three icons: a hamburger menu, a circle, and a back arrow.



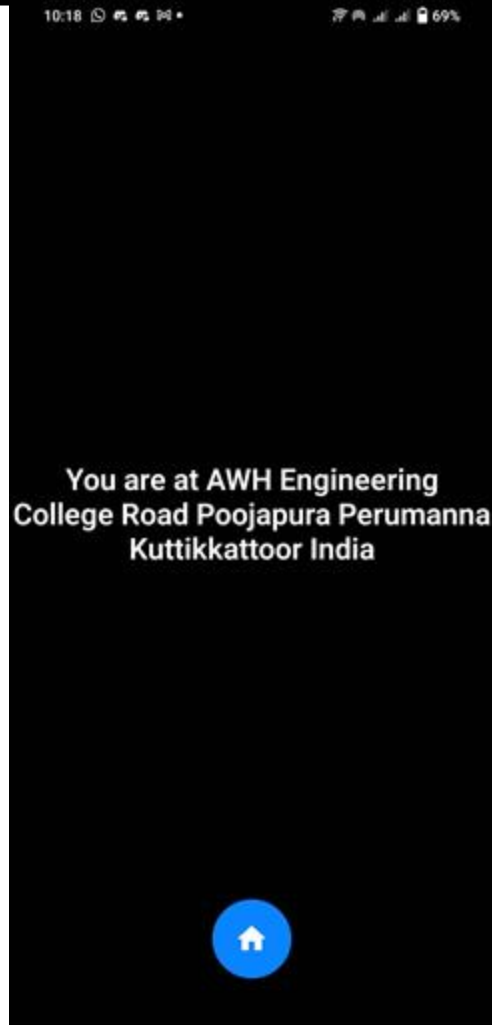
RESULT AND DISCUSSION



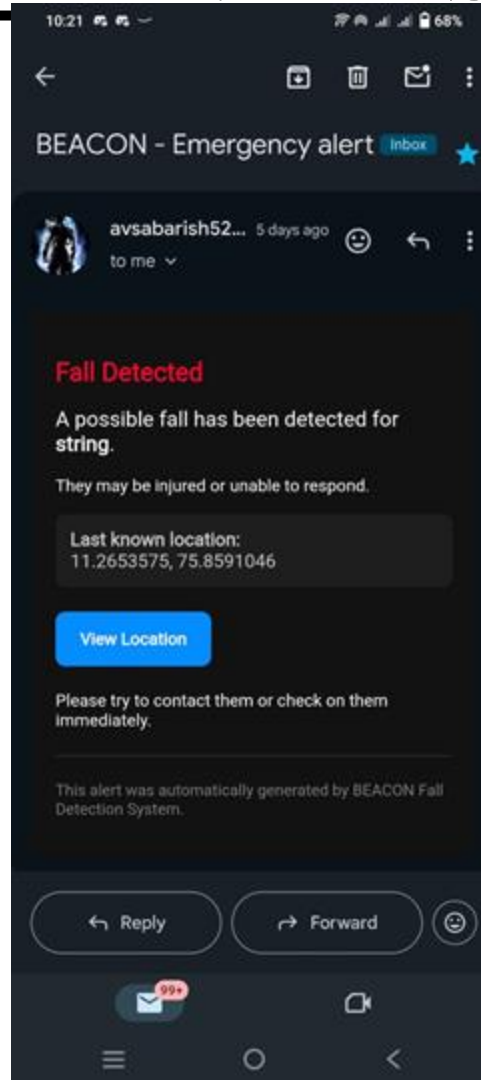
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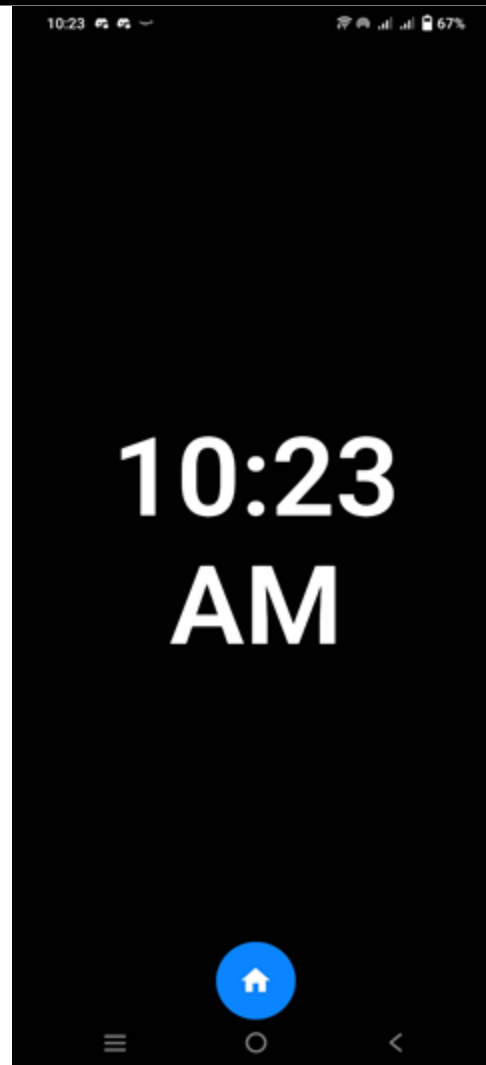
RESULT AND DISCUSSION



RESULT AND DISCUSSION



RESULT AND DISCUSSION



CONCLUSION

- AI-powered assistive cap enhances safety and navigation for visually impaired users.
- Integrates sensors, camera, and edge AI for real-time obstacle and fall detection.
- Improves user independence, awareness, and overall safety.
- Demonstrates effective IoT and mobile app integration.
- Promotes inclusive and accessible smart technology.
- Future scope: GPS navigation, voice commands, cloud analytics, priority obstacles warning, facial recognition.



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THANK YOU

